

MLFF common online-monitor specification

Status: current normative monitoring policy

Architecture: restored-P5 reconciliation 2026-09-14

1. Scope and semantic type boundary

This specification defines deterministic campaign-common **monitoring evidence**. It does not define target-training size, replay-training membership, checkpoint score thresholds, or held-out cross-validation roles.

The current policy families are type-distinct:

OnlineTargetMonitorPolicy
ReplayMonitorPolicy
ResolvedTargetSizePolicy

Their records SHALL remain distinct even when two cardinalities happen to share an integer value.

Online monitors never supply gradients.

2. Common target monitor

OnlineTargetMonitorPolicy owns the common target-monitor subset used by authorized development/model-selection procedures. For **restored current P5**, it is the single target checkpoint/adaptive-stop monitor `M_mon` reused by every selected size, CV fold/seed, and final-production seed/run belonging to the same current campaign/method lineage.

The current requested target-monitor cardinality is exactly:

256 configurations

For restored P5 the realized cardinality is also exactly 256. Fewer than 256 usable eligible configurations is typed P5 infeasibility; there is no current P5 short-parent success state or shrink fallback.

The current deterministic monitor seed is:

161803

The current default strategy identity is:

balanced_condition_run_time_systematic

Changing requested cardinality, seed, strategy, or another policy field changes the policy identity.

Parent role

For restored P5, the target-monitor parent is the neutral label-usable `OUTER_MONITOR` domain only. Gradient-training, held-out CV evaluation, calibration, purge-only, excluded, and locked-test roles are not eligible parents.

Historical/general references to a DATA5 common outer/development-monitor parent are compatibility descriptions; current P5 resolves the neutral parent through the current P1/neutral-substrate authority and does not reactivate a pre-target DATA5 CV owner.

Deterministic selection

The current target-monitor constructor preserves the accepted D2 deterministic construction. At the representation level it records the governed strata/quotas, source/run/time ordering evidence, seed-derived deterministic choices, exact ordered frame/source identities, requested/realized size, policy identity, and label-domain identity required to reconstruct membership.

Every current P5 size/fold/seed/final run using the same compatible campaign monitor identity receives the exact same target-monitor membership/content identity.

Membership before protected-relation qualification

Monitor membership construction and target/monitor protected-relation qualification are different owners and occur in this order:

1. validate the label-usable neutral OUTER_MONITOR parent;
2. sample the exact 256 members under accepted D2;
3. persist/authenticate the immutable monitor record; then
4. use canonical P1 protected-relation authority over the joint monitor/target universe to prove separation from every governed T_N.

A protected-relation conflict fails plan admission. The implementation SHALL NOT relation-prefilter the parent, delete/replace a conflicting selected member, or resample another target monitor.

The selected-only SelectedRelationProjection is insufficient as the sole proof of this cross-role separation because it intentionally excludes frames outside T_selected.

3. Independent true-label replay monitor

ReplayMonitorPolicy owns replay-monitor construction. The current default requested replay-monitor cardinality remains exactly:

512 configurations

The default replay-monitor strategy remains:

chemistry_size_systematic

The true-label replay monitor parent must satisfy the current true-label replay contract. Foundation pseudo labels may be used by a separately identified replay-training path when allowed, but they do not define an absolute true-label replay-validation metric.

Replay-monitor selection uses deterministic ordering across chemistry/composition, atom-count/size grouping, and source order followed by the current systematic selection rule. The materialized replay-monitor artifact is immediately re-inspected; ordered geometry identity, label identity, and true-label mode must match its selection record.

When fewer than 512 eligible true-label replay configurations exist, all eligible configurations are used and the replay-monitor fallback is explicit. This retained replay-monitor behavior does **not** imply a fallback for the restored-P5 256-frame target monitor.

Replay training and replay monitoring are different evidence roles and may not silently alias one another.

4. Record contracts

OnlineTargetMonitorRecord

Binds at least:

- target-monitor role;
- neutral parent-domain identity/digest;
- OnlineTargetMonitorPolicy digest;
- requested size;
- realized size;
- exact ordered selected identities/source indices;
- per-stratum available/selected and other D2 reconstruction evidence;
- strategy and seed;
- label-domain identity; and

- label-usability evidence required by the consuming target metric.

For restored current P5, requested and realized sizes are both exactly 256 and no target-monitor fallback reason can authorize success.

Plan-level protected-relation separation evidence is a descendant of the monitor record and the governed target binding(s); it is not folded into monitor membership identity.

ReplayMonitorRecord

Binds at least:

- replay-monitor role;
- replay source/label lineage;
- ReplayMonitorPolicy digest;
- requested and realized sizes;
- exact ordered selected identities/source indices;
- strategy/seed fields owned by the replay policy;
- true-label mode where required;
- materialized artifact digest where applicable; and
- explicit replay fallback reason where applicable.

For restored P5, the exact common target-monitor record is bound by current CV/final plans descending from PostSelectionMethodIdentity. Broad TrainingProtocolIdentity monitor fields remain applicable only to separately current non-P5 consumers/history and do not create a second P5 owner.

5. Leakage and independence invariants

- Monitoring configurations never contribute gradients.
- The common target monitor is development/model-control evidence, not held-out CV evidence.
- Held-out CV evaluation and locked tests cannot be promoted into online-monitor roles.
- Target-training geometries may not violate governing replay-monitor separation requirements.
- Restored-P5 target-monitor membership is selected once per compatible campaign identity and reused; it is not redrawn per epoch, selected size, fold, seed, or final run.
- Target-monitor membership is not mutated to repair protected-relation conflicts.
- The monitor seed is policy identity, not an informal hint.
- A monitor cardinality is never interpreted as `N_selected`.
- Common target-monitor labels may control checkpoint/adaptive-stop choice and representative target metrics but may not enter foundation-P5 residual-E0 fitting.

6. Relationship to target-size study

The target-size reducer may consume monitor evidence only under its separately accepted P3 method. The target-size study does not own restored-P5 monitor construction and cannot change common-monitor membership between P5 sizes/folds/seeds.

The target-size ladder remains an independent population; monitor sizes 256 and 512 are semantically independent of target-training size.

A P5-only monitor schema/currentness cutover SHALL NOT blanket-invalidate unchanged P3 evidence unless an actual shared P3 monitor semantic changed.

7. Precision and accumulation boundary

Monitor inference uses the learned-model precision/backend declared by the applicable current method/runtime identity. For broad non-P5 DATA8 consumers this may be TrainingProtocolIdentity; for restored P5 it is the

current P5 method/runtime lineage. mdstats-owned metric accumulation remains under the current numerical-precision specification and is not weakened by model dtype.

Changing a numerically material model precision/backend creates a different applicable method/runtime identity where the current architecture declares it method-defining.

8. Persistence and unsupported historical records

Current monitor artifacts are accepted only when their current schema, content, parent lineage, policy digests, and consumer-specific invariants validate.

For restored P5, obsolete fold-local, final-specific, M3-derived, alternate-parent, or short target-monitor records remain historical. Compatibility aliases/deserialization cannot make them current.

A campaign whose required current monitor records cannot validate under the applicable generation requires re-preparation or typed infeasibility; it does not silently reinterpret old monitor semantics.

9. Acceptance requirements

Current monitor qualification covers at least:

1. deterministic target/replay membership regeneration;
2. restored-P5 exact target requested/realized size 256 with no short-parent success;
3. common target-monitor identity across every compatible current P5 size/fold/final/seed plan;
4. accepted condition/run/time or more exact D2 distribution semantics rather than first-N truncation;
5. target-monitor label usability;
6. post-sampling protected-relation failure without filtering/replacement/resampling;
7. proof that selected-only relation projection is not the sole target-vs-monitor separation authority;
8. true-label replay enforcement where required and exact replay artifact round-trip;
9. applicable method/plan identity binding;
10. corruption/staleness rejection; and
11. proof that monitor cardinalities are not consumed as target-size authority.